

Acute Decompensated Heart Failure

Inpatient management for IM residents — profile, decongest, optimize, discharge

Step 1 — Profile: congestion × perfusion

Two bedside axes drive the first orders — not the EF

	Dry	Wet (congested)
Warm	Compensated — optimize GDMT	WARM-WET: IV loop diuretic — the classic admit
Cold	Cautious fluid / inotrope	COLD-WET: diuresis + inotrope/vasodilator; ICU-level

PEARL Hypoperfusion ≠ hypotension: narrow pulse pressure, cool extremities, ↑ lactate/Cr, somnolence.

Congestion vs perfusion — the signs

WET (congestion)

- JVD, orthopnea, PND
- Peripheral edema, weight gain
- B-lines / IVC plethora on POCUS
- S3 gallop

COLD (hypoperfusion)

- Narrow pulse pressure (<25% SBP)
- Cool, mottled extremities
- Rising lactate and creatinine
- Somnolence, low urine output

PEARL Most ward HF admits are warm-and-wet → diuresis alone. Cold-and-wet → higher acuity.

Step 2 — Decongest with IV loop diuretics

Dose by home dose, confirm by natriuretic response (DOSE trial)

- Give IV, not PO — gut edema cripples oral absorption
- Start IV furosemide $\approx 1\text{--}2.5\times$ total home daily PO dose, IV BID (naive: 40–80 mg)
 - Reassess at 2 h: spot urine Na $>50\text{--}70$ mmol/L or UOP $>100\text{--}150$ mL/h = adequate
 - If inadequate — double the dose now; don't wait a day
- Resistant: add thiazide (metolazone / IV chlorothiazide); consider acetazolamide (ADVOR), early SGLT2i
- Daily weights, strict I/O, BMP q12–24h; target net $-1\text{--}2$ L/day

PEARL A rising Cr during effective diuresis in a still-wet, urine-making patient is usually hemoconcentration — keep going.

Step 3 — Start GDMT before discharge

The four pillars of HFrEF — the admission is the intervention

Pillar	Agent	Inpatient note
ARNI / ACEi / ARB	sacubitril-valsartan	Hold SBP<90, K>5.5, AKI; 36 h ACEi washout
Beta-blocker	carvedilol, metoprolol succ.	Continue if warm; don't start until euvolemic & off inotropes
MRA	spironolactone, eplerenone	K<5.0, eGFR>30
SGLT2 inhibitor	dapagliflozin, empagliflozin	Start in-hospital — safe, speeds decongestion (EMPULSE)

PEARL STRONG-HF: rapid in-hospital initiation + up-titration with close follow-up improves outcomes.

Find the precipitant — FAILURE

- F — Forgot meds / Fluid & salt excess
- A — Arrhythmia (AF with RVR)
- I — Ischemia / Infection
- L — Lifestyle (dietary sodium, alcohol)
- U — Upregulation (thyroid, anemia, pregnancy)
- R — Renal failure / Rx (NSAIDs, non-DHP CCB, TZD)
- E — Embolus (pulmonary embolism)

PEARL Work-up: ECG + troponin, BNP/NT-proBNP, CBC, BMP, TSH, CXR; echo if no recent EF.

Discharge bundle — cut the readmission

DO

- Transition IV→PO diuretic, observe 24 h euvolemic
- Document EF, dry weight, diuretic dose, 4 pillars
- Teach daily weights & low-Na diet
- Arrange 7-day follow-up

AVOID

- Discharging still-congested ('looks better')
- NSAIDs, TZDs, non-DHP CCB in HFrEF
- Starting/up-titrating BB while wet or cold
- No follow-up scheduled

PEARL Residual congestion at discharge is a leading driver of 30-day readmission.

Take-home points

- Profile on two axes (wet/dry × warm/cold) before the EF
- IV loops dosed to a measured 2-h natriuretic response; double if short
- Don't stop diuresis for a small Cr bump in a still-wet patient making urine
- Start all four GDMT pillars in-house; hold only BB initiation while wet/cold
- Discharge euvolemic, precipitant named, GDMT documented, 7-day follow-up